



Numbers and Numerals

A **number** is an item that describes a magnitude or a position.

Numbers are classified into two types, namely cardinal numbers and ordinal numbers.

Cardinal numbers are numbers which allow us to count the objects or ideas in a given collection. Example, 1,2,3...,1000, 100000 while **ordinal numbers** state the position of the individual objects in a sequence. Example, First, second, third...

Numerals are symbols, or combination of symbols which describe a number.

The most widely used numerals are the Arabic numerals and the Roman numerals.

Arabic numerals were simply the modification of the Hindu-Arabic number signs and are written in Arabic digits. Taken singly, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and in combination 20, 21, 22, ... 1999, ...

The Roman numerals are numbers which are written in Latin alphabet. Example MCMXCIV.

The following are Roman numerals and their equivalent Arabic numbers:

I = 1	L = 50	M = 1000
V = 5	C = 100	
X = 10	D = 500	

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The Romans used the following to indicate large numbers:

1. Bracket - to multiply it by 100 times.

$$|V| = 500$$

2. Vinculum (bar above the number) – to multiply the number 1000 times.

$$\bar{V} = 5,000$$

3. Doorframe – to multiply the number by 1000000 times

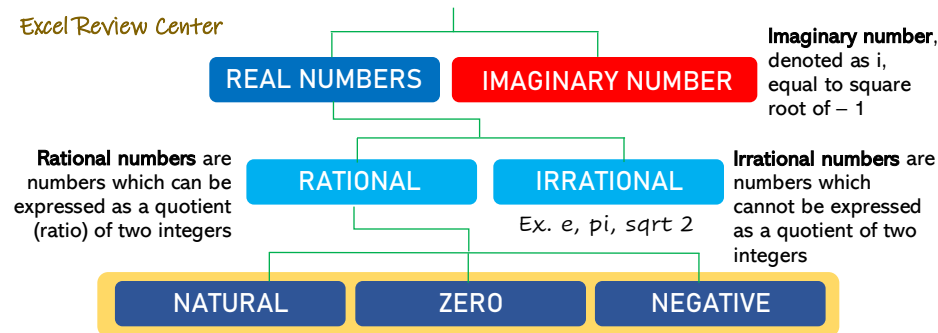
$$|V| = 5,000,000$$

A **digit** is a specific symbol or symbols used alone or in combination to denote a number.

For example, the number 21 has two digits, namely 2 and 1. In Roman numerals, the number 9 is denoted as IX. So the digits I and X were used together to denote one number and that is the number 9. In mathematical computations or in some engineering applications, a system of numbers using cardinal numbers was established and widely used.

System of Numbers

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Integers are all the natural numbers, the negative of the natural numbers and the number zero.

Natural numbers are numbers which are considered as the "counting numbers". Ex. 1, 2, 3, ...



For non-terminating decimals:

If it is repeating, like 0.3333..., it is **rational number**

If it is non-repeating, like $\pi = 3.14159...$, it is **irrational number**

Complex Numbers

A **complex number** is an expression of both real and imaginary number combined. It takes the form of $a + bi$, where "a" and "b" are real numbers. If $a = 0$, then **pure imaginary number** is produced while **real number** is obtained when $b = 0$.

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Imaginary number and its equivalent:

$$i = \sqrt{-1} \quad i^2 = -1 \quad i^3 = -i = -\sqrt{-1} \quad i^4 = 1$$

Properties of INTEGERS

ADDITION PROPERTIES

1. Closure property

$$a + b = \text{integer}$$

2. Commutative property

$$a + b = b + a$$

3. Associative property

$$(a + b) + c = a + (b + c)$$

4. Identity property

$$a + 0 = a$$

0 is called the **additive identity**

5. Inverse property

$$a + (-a) = 0$$

$-a$ is called the **additive inverse**

6. Distributive property

$$a(b + c) = ab + ac$$

MULTIPLICATION PROPERTIES

1. Closure property

$$ab = \text{integer}$$

2. Commutative property

$$ab = ba$$

3. Associative property

$$(ab)c = a(bc)$$

4. Identity property

$$a + 1 = a$$

1 is called the **multiplicative identity**

5. Inverse property

$$a \left(\frac{1}{a} \right) = 1$$

$\frac{1}{a}$ is called the **multiplicative inverse**

6. Distributive property

$$a(b + c) = ab + ac$$

7. Multiplication property of zero

$$a(0) = 0$$

EQUALITY PROPERTIES

1. Reflexive property

$$a = a$$

2. Symmetric property

$$\text{If } a = b, \text{ then } b = a$$

3. Transitive property

$$\text{If } a = b \text{ and } b = c, \text{ then } a = c$$

4. Substitution property

If $a = b$, then a can be replaced by b in any expression involving a

5. Addition / Subtraction property

$$\text{If } a = b, \text{ then } a + c = b + c$$

$$\text{If } a = b, \text{ then } a - c = b - c$$

6. Multiplication / Division property

$$\text{If } a = b, \text{ then } ac = bc$$

$$\text{If } a = b, \text{ then } \frac{a}{c} = \frac{b}{c}, c \neq 0$$

7. Cancellation property

$$\text{If } a + c = b + c, \text{ then } a = b$$

$$\text{If } ac = bc \text{ and } c \neq 0, \text{ then } a = b$$

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